PRODUCT REQUIREMENTS DOCUMENT

Scout AI - Autonomous Deep Research Engine

A Streaming, Evidence-Grounded Multi-Agent Research Platform

Built on FastAPI, DeepAgents, LangGraph, LangChain, Tavily, OpenAI/Nebius, and LangSmith

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1. Executive Summary

Scout AI is a browser-based autonomous deep-research application that converts a natural-language question into a sourced research dossier. A DeepAgents supervisor delegates evidence acquisition, analysis, and publication to three bounded sub-agents. Each stage works from an immutable brief and exchanges compact Markdown artifacts through exact virtual paths: /brief.md, /research.md, /analysis.md, and /final.md.

The product supports live research through Tavily plus an OpenAI-compatible model provider, and an explicitly labeled simulation mode that previews workflow structure without claiming verified findings. Live execution is streamed as LangGraph events rather than awaited as one opaque result, allowing the UI to show real milestones, agent activity, retries, backoff, model turns, prompt-token telemetry, and terminal errors.

The implemented application is a single-process FastAPI service with process-local active workers and JSON-backed history. That architecture is appropriate for demonstration and controlled local use, but it does not yet provide durable graph checkpointing, distributed job execution, authentication, encrypted multi-user secret management, or restart-safe task continuation. This PRD distinguishes those current facts from target requirements and roadmap items.

> **Product Promise:** Answer the user's exact question with current, decision-relevant evidence; preserve traceability from recommendation to source; and make multi-agent progress understandable and honest.

2. Goals & Success Criteria

2.1 Product Goals (In Scope)

1. Produce a complete executive research dossier from one user query without requiring the user to coordinate individual agents.
2. Ground conclusions in authoritative URLs and explicit evidence IDs rather than relying on model memory alone.
3. Keep consumer-product research current through runtime dates, a one-year source window, market context, and a latest-available-for-purchase requirement.
4. Bound latency, cost, recursion, model turns, retries, search fan-out, and inter-stage context size.
5. Expose honest stage milestones and live activity instead of continuously increasing synthetic progress.
6. Derive dashboard metrics from the current task's /research.md, /analysis.md, and /final.md artifacts.
7. Surface the failing agent's name and actionable reason when a run cannot complete.

2.2 Out of Scope

8. Guaranteed truth or completeness for every open-web topic; Scout AI provides evidence-backed research, not certainty.
9. Autonomous purchasing, trading, legal filing, medical diagnosis, or other transaction execution.
10. Multi-tenant enterprise security, RBAC, SSO, or durable distributed task processing in the current local architecture.
11. Full-document retrieval and extraction from every search result; the current research tool passes compact Tavily snippets and metadata.
12. Automatic recovery of an in-flight graph after an application restart; current startup reconciliation marks interrupted work failed.

2.3 Success Metrics

Metric	Target	Measurement
Topic fidelity	100% of completed artifacts repeat and answer the exact original query	Blocking artifact validator
Source traceability	100% of research evidence records include a URL	research.md schema and URL validation
Hard-path compliance	100% of agent file operations use approved virtual paths	SharedWorkspacePathMiddleware logs/tests
Terminal-state freshness	UI reflects completed/failed state within 2 seconds	SSE integration test and browser telemetry
Research completion rate	>= 90% for supported benchmark queries	LangSmith evaluation dataset
Latency	P50 < 180s; P95 < 360s for comprehensive live research	Task duration telemetry
Citation usefulness	>= 85% of sampled citations support the associated claim	Human/LLM-judge audit
Constraint compliance	100% adherence to configured turn, recursion, concurrency, and artifact budgets	Runtime middleware + regression tests

3. Target Users, Actors & Personas

Actor / Persona	Primary Need	Scout AI Value	Key Concern
Executive decision-maker	A concise decision brief with risks and actions	Executive synthesis, comparative matrices, citations	Confidence and source quality
Strategy or market analyst	Rapid evidence collection and structured comparisons	Evidence ledger, analysis matrix, downloadable artifacts	Coverage gaps and reproducibility
Consumer / procurement researcher	Current product recommendations and ownership trade-offs	Runtime date, one-year search window, latest-purchase requirement	Model-year and availability accuracy
Product manager	Competitive, customer, and technology research	Reusable query starters, radar/SWOT, artifact history	Metrics must reflect the current query
Research lead / reviewer	Inspectable provenance and assumptions	research.md, analysis.md, final.md, LangSmith traces	Hallucination and unsupported synthesis
Application operator	Configure providers and diagnose failures	Settings UI, agent names in errors, live telemetry	Timeouts, quota, TLS, restart safety
API integrator	Programmatic task initiation and event consumption	REST start/history/export endpoints and SSE stream	Stable contracts and authentication

3.1 Stakeholders

13. Product owner: defines supported research experiences, reliability bar, and launch scope.
14. AI/agent engineer: owns prompts, graph behavior, middleware, provider/model routing, and evaluations.

15. Backend engineer: owns FastAPI contracts, persistence, task lifecycle, export, and operational safety.
16. Frontend engineer/designer: owns research intake, live topology, progress, dossier visualization, and accessibility.
17. Security/operations reviewer: owns key handling, network trust, auditability, deployment, and incident response.

4. End-to-End User Journey

18. Open Scout AI and begin from a clean dashboard with no stale query, progress, results, or metrics.
19. Enter a specific research question; choose depth (quick, comprehensive, exhaustive), focus, final synthesis model, and live or simulation mode.
20. Submit the query. The API validates non-empty input and required provider keys, creates a task, and returns a task ID.
21. Scout AI constructs /brief.md with the exact question, depth/focus, runtime date, one-year source-window start, United States market, and latest-purchasable-product requirement.
22. The Orchestrator reads the brief and delegates Research, Analysis, and Writer work using full task descriptions and exact shared paths.
23. The UI receives SSE events every approximately 600 ms and shows milestone progress, the active green topology path, logs, model turns, attempts, prompt tokens, and last activity only when available.
24. On success, Scout AI validates artifacts, derives current-task metrics, persists history, renders the Markdown dossier and charts, and enables DOCX export.
25. On failure, Scout AI stops at 100%, identifies the responsible agent where known, records the actionable error, and leaves the task in history.
26. The user can inspect research, analysis, final report, workspace files, citations, and historical dossiers or start a clean new task.

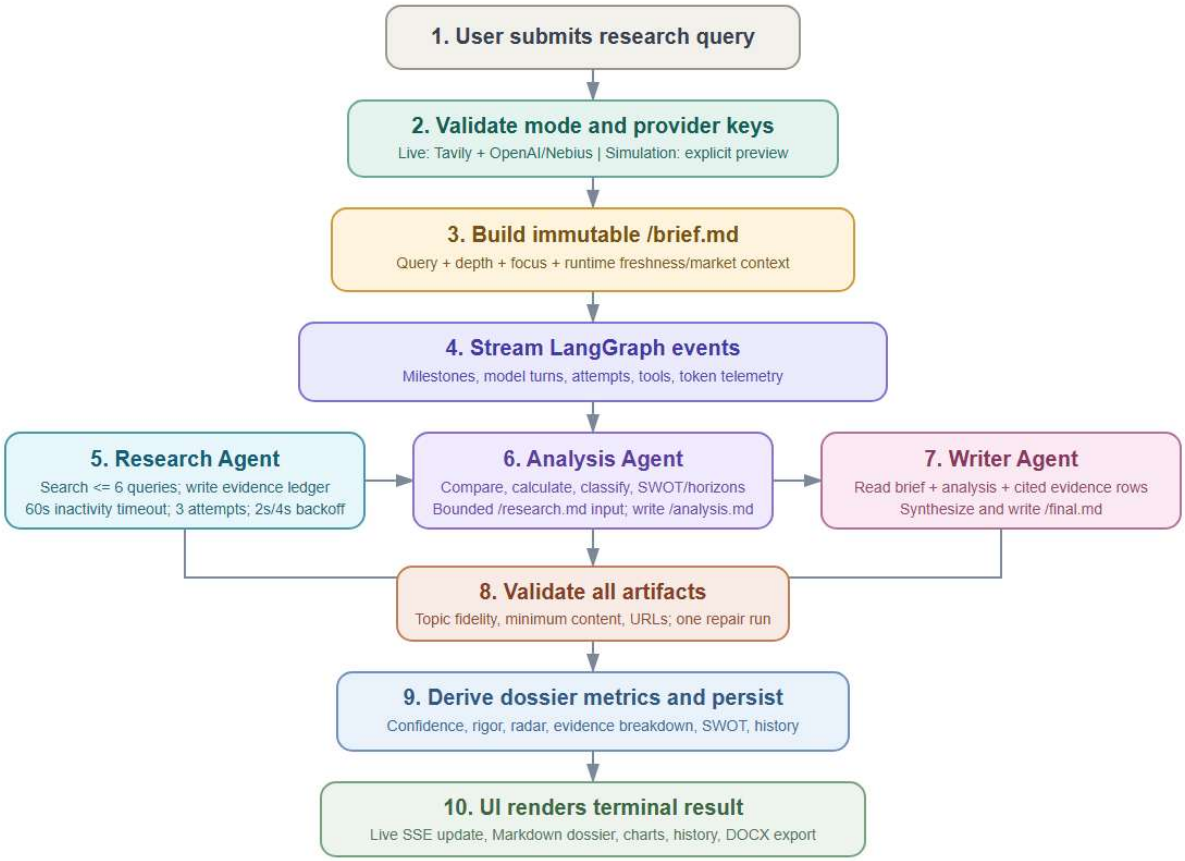


Figure 1. Scout AI end-to-end live research and artifact flow.

5. Functional Requirements

ID	Capability	Requirement / Acceptance
FR-001	Query intake	Accept a non-empty natural-language research query; reject blank input with HTTP 422.
FR-002	Research controls	Support quick/comprehensive/exhaustive depth, general/news/finance/academic focus, model selection, and explicit simulation mode.
FR-003	Clean start	On page refresh, clear the active query, progress, results, activity labels, and derived metrics.
FR-004	Live prerequisites	Require Tavily plus either OpenAI or Nebius credentials for live mode; return HTTP 503 when unavailable.
FR-005	Immutable brief	Create /brief.md for every run and make it the authoritative context for all agents.
FR-006	Runtime freshness	Write AS_OF_DATE, SOURCE_WINDOW_START, MARKET, and PRODUCT_REQUIREMENT into /brief.md.
FR-007	Exact delegation context	Every task delegation includes the complete original query, stage objective, inputs, output path, and acceptance check.
FR-008	Research stage	Search authoritative web sources in bounded parallel batches and write a compact evidence ledger with URLs to /research.md.
FR-009	Analysis stage	Read bounded evidence, classify facts/inferences/assumptions, compare options, calculate metrics, and write /analysis.md.
FR-010	Writer stage	Use /brief.md, /analysis.md, and cited evidence excerpts to create a publication-grade /final.md.

ID	Capability	Requirement / Acceptance
FR-011	Artifact validation	Block completion when artifacts are missing, too short, off-topic, placeholder-based, or lack research URLs; permit one repair workflow.
FR-012	Event streaming	Use LangGraph astream_events(version='v2') and SSE; do not await one opaque agent result.
FR-013	Honest progress	Use milestones (Research 35-55%, Analysis 60-75%, Writer 80-92%, completion 100%) with activity telemetry.
FR-014	Retries	Apply one initial attempt plus two retries to Research, Analysis, and Writer agents after 60 seconds of inactivity.
FR-015	Backoff	Wait 2 seconds after the first timeout and 4 seconds after the second before retrying.
FR-016	Bounded execution	Enforce agent model-call limits, graph recursion limit, and maximum concurrency.
FR-017	Context budgets	Cap intermediate artifacts, line reads, result counts, and snippets to prevent context growth.
FR-018	Failure attribution	Include the failing agent name for timeout, iteration, and graph-stop failures when identifiable.
FR-019	Artifact-derived analytics	Populate SWOT/horizon, evidence rigor, radar, logic rigor, read time, and strategic confidence from the current artifacts.
FR-020	History	List, load, recompute metrics for, and delete stored dossiers.
FR-021	Export	Export final report and optional research/analysis appendices as a DOCX download.
FR-022	Settings	Configure API keys, router/synthesis models, base URL, LangSmith key, and LangSmith project through the UI/API.
FR-023	TLS trust	Use the Windows trust store for Tavily, LangSmith, and OpenAI-compatible HTTPS connections.
FR-024	Simulation labeling	Never represent simulation output as verified evidence; label all preview artifacts and metrics accordingly.

6. Research Brief & Artifact Contracts

6.1 Immutable Research Brief

The brief prevents context loss across isolated sub-agent sessions. It is generated once per task, injected into the DeepAgents virtual filesystem, and read before every stage.

```
ORIGINAL USER QUESTION: <verbatim query>
RESEARCH DEPTH: <quick|comprehensive|exhaustive>
RESEARCH FOCUS: <general|news|finance|academic>
AS_OF_DATE: YYYY-MM-DD
SOURCE_WINDOW_START: YYYY-MM-DD (AS_OF_DATE - 365 days)
MARKET: United States
PRODUCT_REQUIREMENT: Latest model/version currently available for purchase
```

6.2 Shared Artifact Contracts

Path	Owner	Purpose	Implemented Budget
/brief.md	Application / Orchestrator	Immutable query and runtime context	Read <= 60 lines
/research.md	Research Agent	Evidence Ledger: <= 30 records plus <= 5 gaps	<= 100 lines and <= 16,000 chars

Path	Owner	Purpose	Implemented Budget
/analysis.md	Analysis Agent	Decision summary, finding matrix, comparison, risks, actions, SWOT/horizons	<= 80 lines and <= 12,000 chars
/final.md	Writer Agent	Publication-grade answer and citations	Orchestrator final read <= 120 lines

> **Context quarantine:** Sub-agents do not inherit the prior user conversation. /brief.md plus the full task description are the only authoritative bridges for the original topic.

7. System Architecture Overview

Scout AI is a layered local web application. The browser submits and visualizes work; FastAPI manages task state and streaming; the DeepAgents supervisor executes a LangGraph-backed agent graph; specialized agents use a virtual shared workspace; external model/search providers supply inference and current web evidence; LangSmith receives nested traces; and a JSON file stores dossier history.

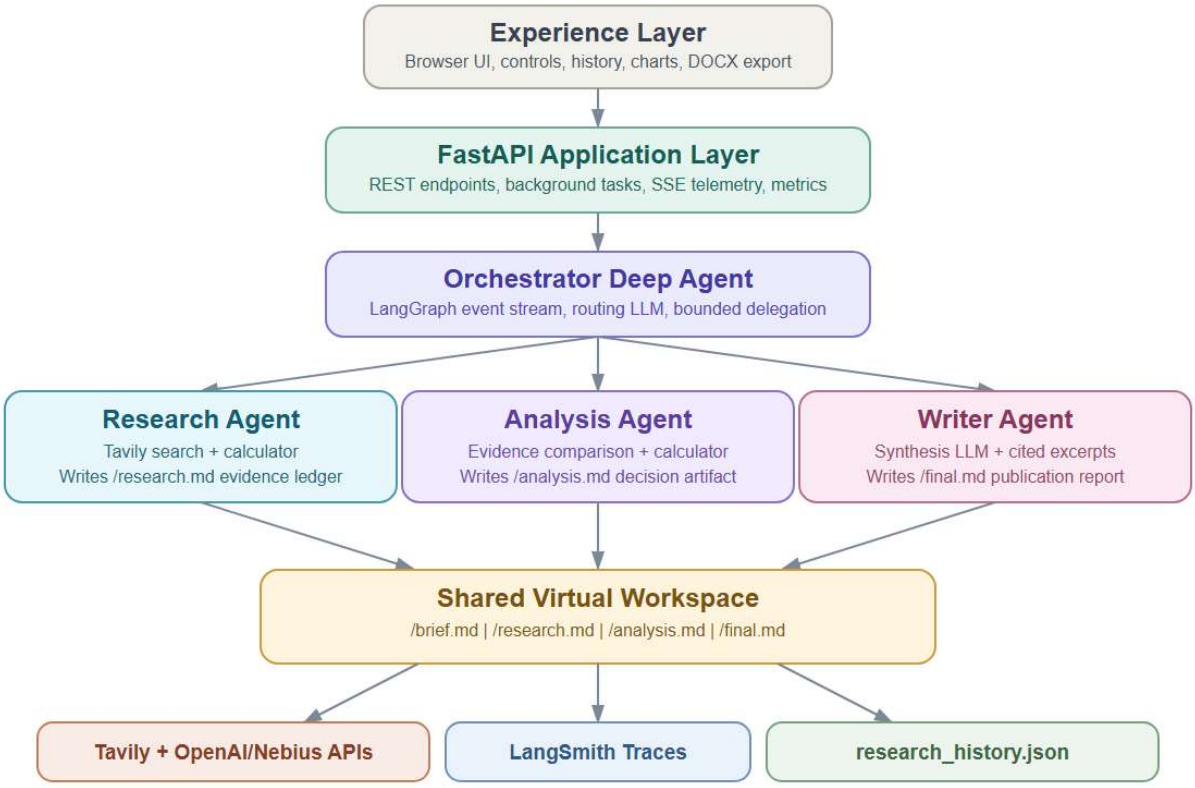


Figure 2. Implemented Scout AI component and data architecture.

7.1 Runtime Boundaries

27. Browser boundary: static HTML/JavaScript, Tailwind styling, event-stream consumer, charts, Markdown/KaTeX rendering, settings, history, and export actions.
28. Application boundary: one Uvicorn/FastAPI process, in-memory TASKS dictionary, FastAPI BackgroundTasks, JSON persistence, and direct provider network calls.

29. Agent boundary: a DeepAgents supervisor compiled over LangGraph; sub-agents have isolated prompts/tools and share virtual FileData objects.
30. Provider boundary: Tavily search and OpenAI-compatible chat completions through OpenAI or Nebius endpoints.
31. Observability boundary: standard LangSmith environment variables automatically capture nested model, tool, and sub-agent runs.

7.2 Current Persistence Model

Completed and failed dossier records are saved to research_history.json (latest 50 tasks). Active work is not checkpointed outside the process. On application startup, any persisted task still marked in_progress is reconciled to failed with an explicit restart-interruption error.

8. Agent Architecture & LLM Routing

8.1 Agent Roster

Agent	Responsibility	Tools	Output	Turn Limit
Orchestrator	Plan, delegate, review bounded artifacts, validate, repair once, terminate	internet_search, calculator, DeepAgents task/file tools	Final graph state	12
Research Agent	Acquire current authoritative evidence and source metadata	internet_search, calculator, read_file, write_file	/research.md	10
Analysis Agent	Compare evidence, calculate, classify, identify risks/actions/SWOT	calculator, read_file, write_file	/analysis.md	7
Writer Agent	Synthesize decision-ready report from brief, analysis, and cited rows	read_file, grep, write_file	/final.md	10

8.2 LLM Role Routing

Scout AI resolves two logical model roles. The router model is used by the Orchestrator, Research Agent, and Analysis Agent for planning, tool selection, and structured reasoning. The synthesis model is reserved for the Writer Agent. When both roles resolve to the same model, one model client is reused.

Model / Family	Provider Path	Role / Availability	Notes
Qwen/Qwen3-30B-A3B-Instruct-2507	Nebius OpenAI-compatible API	Default Nebius router and synthesis; settings option	Current local baseline; selected when OpenAI key is absent
Qwen/Qwen3-235B-A22B-Instruct-2507	Nebius OpenAI-compatible API	Optional final synthesis	Higher-quality/larger writer choice exposed in settings
gpt-4o-mini	OpenAI API	Default OpenAI router and synthesis	Fast/cost-efficient default when OpenAI is configured
gpt-4o	OpenAI API	Optional final synthesis	Settings option for higher-precision general synthesis
o3-mini	OpenAI API	Optional final synthesis	Settings option oriented to reasoning/STEM
o1	OpenAI API	Optional final synthesis	Settings option for deeper reasoning
claude-3-7-sonnet	Catalog entry only	Returned by /api/config catalog; not in settings selector	No native Anthropic client is implemented; requires a compatible endpoint before production use

> **Model configuration:** All model clients use temperature 0.1, a 60-second provider request timeout, and two provider-level retries. Agent-level inactivity retries are a separate middleware layer.

8.3 Delegation Contract

```
ORIGINAL USER QUESTION: <verbatim question from /brief.md>
STAGE OBJECTIVE: <specific bounded work>
REQUIRED INPUTS: </brief.md and approved prior-stage paths>
REQUIRED OUTPUT: </research.md | /analysis.md | /final.md>
ACCEPTANCE CHECK: Output directly addresses the original question
```

9. Guardrails & Grounding Design

Control	Implementation	Risk Addressed
Exact-query repetition	Every stage must begin with the exact original question	Topic drift across isolated contexts
Research URL requirement	research.md must contain at least one HTTP(S) source URL	Untraceable model-only research
Evidence ledger	Compact rows with ID, claim/metric, date/scope, URL, type, confidence	Verbose unstructured handoffs
Fact/inference/assumption split	Analysis finding matrix labels epistemic status and evidence IDs	Inference presented as fact
Relevant excerpt retrieval	Writer greps cited evidence IDs; bounded fallback only when IDs are absent	Context inflation and unrelated evidence
Freshness context	Runtime date, one-year window, U.S. market, latest-purchase requirement	Stale consumer-product recommendations
Tavily time range	internet_search sets time_range='year'	Older result ranking
Artifact caps	Hard line/character budgets with cap notice	Runaway prompts and provider latency
Tool safety	Calculator accepts only digits and standard arithmetic characters	Arbitrary code execution
Path enforcement	Fixed workspace files and bounded read/write middleware	Uncontrolled filesystem exploration
Final validation	Minimum length, exact query, URL, and placeholder checks; one repair attempt	False success with incomplete artifacts

9.1 Known Grounding Limits

32. A one-year search filter constrains result recency but does not prove that a product model is currently purchasable or that a cited model year is the newest available.
33. The Tavily tool returns title, URL, compact snippet, published date, and score; it does not fetch and archive full source documents.
34. Search snippets can omit qualifiers, paywalled details, methodology, or regional availability; key decisions still require primary-source verification.
35. Strategic confidence is a deterministic artifact-quality score, not a calibrated probability that every conclusion is correct.

10. State, Persistence & Task Lifecycle

State Area	Current Storage	Lifecycle / Semantics
Active task object	Process-local TASKS dictionary	Created at POST /start; mutated by BackgroundTasks; lost if process exits
Agent graph state	In-memory LangGraph execution	Streamed through astream_events; no durable checkpointer configured
Shared artifacts	DeepAgents virtual FileData during execution; copied into task record on completion	brief/research/analysis/final exact paths
History	research_history.json	Latest 50 records; saved repeatedly; reloaded on startup
Interrupted work	Startup reconciliation	Persisted in_progress task becomes failed with restart-interruption message
Provider settings	.env plus process environment	Updated through settings API; keys are not returned to the browser
Trace history	LangSmith project when configured	Nested agent/model/tool runs and metadata

10.1 Task Status Model

```
in_progress -> completed
in_progress -> failed
in_progress --application restart--> failed (startup reconciliation)
completed/failed -> history load or delete

There is no current paused, cancelled, queued, or resumable status.
```

10.2 Target Durability Requirement

A production release should move execution into durable workers backed by a database and LangGraph checkpointer. The task ID must remain stable across process restarts; every stage write and terminal transition must be idempotent; active work must either resume from the last checkpoint or fail with a durable, visible reason.

11. Error Handling & Recovery

Failure	Detection	Current Recovery / Surface
Missing query	Input validation	HTTP 422 with specific message
Missing live credentials	Start endpoint provider check	HTTP 503; user may explicitly choose simulation
TLS certificate chain	HTTPS client exception	truststore injects Windows certificate store; verification remains enabled
Tavily transient failure	Search exception	Up to 3 total attempts with 2s/4s waits
Sub-agent inactivity	No model/tool/token/chain callback for 60s	Cancel attempt; retry twice with 2s/4s backoff; name failing agent
Model-call limit	ModelCallLimitMiddleware	Raise agent-specific iteration-limit error
Graph stop failure	LangGraph GraphRecursionError	Surface active agent plus configured 50-step graph budget
Provider quota/credits	429/quota/credit error text	Actionable billing/provider error

Failure	Detection	Current Recovery / Surface
Artifact validation	Blocking post-run validator	One bounded repair graph; then fail explicitly
Application restart	Startup sees persisted in_progress task	Mark failed; current execution cannot resume
SSE disconnect	Browser EventSource.onerror	Current UI closes stream and re-enables start; target behavior is reconnect/fetch terminal state

> **Reliability distinction:** The 60-second policy measures inactivity, not total stage duration. Healthy model tokens, tool calls, and chain events reset the clock so productive long-running work can continue.

12. Observability, Streaming & LangSmith

12.1 Event Streaming

The graph is executed with agent.astream_events(..., version='v2'). Nested task ancestry attributes model and tool events to the correct sub-agent. The backend converts events into StreamTelemetry and publishes current task state through an SSE endpoint roughly every 600 ms.

Telemetry	Examples	UI / Operational Use
Milestone	30 orchestrator; 35/55 research; 60/75 analysis; 80/92 writer; 100 terminal	Honest progress bar
Agent activity	active_agent, current_tool, last_activity_epoch	Green topology path and slow-provider warning
Model usage	turn number/limit; prompt/completion/total tokens when available	Cost/context diagnostics
Retry state	attempt, max_attempts, timeout, backoff, activity_state	Visible retry countdown and failure context
Delegation	per-agent invocation counts	Detect rework and repeated stages
Artifacts	current file lengths; full files at terminal state	Verify stage output and render result
Quality	source count, confidence, metrics	Dossier analytics

12.2 LangSmith Tracing

36. LANGSMITH_API_KEY enables tracing; LANGSMITH_TRACING defaults true when the key is present.
37. LANGSMITH_PROJECT defaults to scout-ai and is configurable in settings.
38. Standard LangChain/DeepAgents integration emits nested model, tool, chain, and sub-agent runs.
39. Trace config includes task ID, depth, focus, selected model, execution mode, recursion limit, and tags.
40. Recommended evaluations: groundedness, citation support, topic fidelity, route accuracy, artifact contract, latency, token usage, and completion rate.

12.3 Artifact-Derived Dashboard Metrics

SWOT and horizons are extracted from /analysis.md; evidence rigor from /research.md; comparative radar from structured analysis; read time from /final.md; and strategic confidence from current-task evidence coverage, source quality, analysis grounding, gap disclosure, freshness, and final traceability. The UI must not substitute placeholder values when artifacts are absent.

13. UI/UX Requirements

Area	Requirement
Research intake	Prominent query field, depth controls, focus dropdown defaulting to All Authoritative Web, explicit simulation toggle, and start action.
Clean refresh	Clear query, progress, results, activity telemetry, charts, confidence, and selected dossier; preserve only configuration/history.
Live topology	Distinct high-contrast node colors; active path green; readable labels above nodes; clear spacing to Shared Workspace.
Progress	Milestone percentage plus active agent; no synthetic continuous movement.
Optional activity	Hide Model call, Attempt, Prompt tokens, and Last activity fields when their values are absent or waiting.
Latency warning	After 45 seconds without model activity, state that provider response is slow and show estimated timeout remaining.
Backoff	Show agent attempt timed out and exact retry wait (2 or 4 seconds).
Results	Tabs for analytics, final.md, research.md, analysis.md, and shared files.
Analytics	Render current-query strategic confidence, evidence rigor chart, comparative radar, SWOT, and decision horizons.
Accessibility	High contrast, semantic labels, keyboard focus, responsive layout, readable chart legends, and non-color-only status cues.
History	Searchable drawer; status, query, duration, sources, confidence; load and delete actions.
Failure	Stop activity indicator, restore start button, show terminal error and failed agent, and retain the failed task in history.

13.1 Visual System

The interface uses a dark slate background with teal/cyan/indigo accents, high-contrast glass panels, Inter and JetBrains Mono typography, and distinct colors for Orchestrator, Research, Analysis, Writer, and Shared Workspace nodes. Chart palettes must use multiple distinguishable colors and textual legends.

14. Technical Stack Summary

Concern	Implemented Technology	Version / Notes
Language/runtime	Python	Local virtual environment based on Python 3.14
Web API	FastAPI	0.141.1
ASGI server	Uvicorn	0.52.4
Agent framework	DeepAgents	0.7.11
Agent/tool primitives	LangChain + langchain-openai	1.3.18 / 1.6.0
Graph runtime	LangGraph	1.2.11; event streaming v2
Tracing/evals	LangSmith	0.11.2
Web research	Tavily Python SDK	0.8.0

Concern	Implemented Technology	Version / Notes
Configuration	python-dotenv	1.2.3; .env and process environment
Document export	python-docx	1.2.0
TLS trust	truststore	0.10.4; Windows trust store injection
Frontend	HTML5 + vanilla JavaScript	Server-rendered static template; no build step
Styling	Tailwind CSS CDN + custom CSS	CDN dependency currently unpinned
Visualization	Chart.js	CDN; radar and evidence-rigor charts
Markdown/math	marked.js + KaTeX	CDN-based rendering
Icons/animation	Font Awesome + canvas-confetti	CDN-based UI enhancement
Persistence	JSON file + process memory	research_history.json; latest 50 tasks

15. Configurable Parameters

15.1 Runtime Reliability and Context Parameters

Parameter	Current Value	Purpose
AGENT_TIMEOUT_SECONDS	60	Inactivity timeout for delegated agents and provider request timeout
AGENT_MAX_RETRIES	2	Two retries after the initial attempt; three attempts total
Retry backoff	2s, then 4s	Computed as 2^attempt_number; not separately configurable
RESEARCH_AGENT_MAX_TURNS	10	Research model-call run limit
ANALYSIS_AGENT_MAX_TURNS	7	Analysis model-call run limit
WRITER_AGENT_MAX_TURNS	10	Writer model-call run limit
ORCHESTRATOR_MAX_TURNS	12	Supervisor model-call run limit
AGENT_MAX_CONCURRENCY	5	LangGraph maximum concurrent work
GRAPH_RECURSION_LIMIT	50	Graph node-transition budget passed at invocation
RESEARCH_ARTIFACT_MAX_LINES	100	Maximum research.md lines
RESEARCH_ARTIFACT_MAX_CHARS	16,000	Hard research.md character cap
ANALYSIS_ARTIFACT_MAX_LINES	80	Maximum analysis.md lines
ANALYSIS_ARTIFACT_MAX_CHARS	12,000	Hard analysis.md character cap
BRIEF_READ_MAX_LINES	60	Maximum brief read
FINAL_READ_MAX_LINES	120	Maximum orchestrator final verification read
WRITER_RESEARCH_FALLBACK_MAX_LINES	40	Fallback evidence read when analysis IDs are missing
SEARCH_MAX_RESULTS_PER_QUERY	3	Maximum Tavily results returned per query
SEARCH_RESULT_SNIPPET_CHARS	600	Maximum snippet characters per result
Research search count	<= 6	Prompt-level total query cap in at most two parallel batches

Parameter	Current Value	Purpose
Tavily time_range	year	One-year search-result constraint
Model temperature	0.1	Low-variance model configuration
SSE interval	~0.6 seconds	Backend event-generator update cadence
Slow-provider warning	45 seconds	Frontend warning threshold before 60-second inactivity timeout
History retention	50 tasks	Latest records persisted to JSON

15.2 Environment and Provider Parameters

Environment Variable	Default / Behavior	Used By
OPENAI_API_KEY	Optional; takes precedence over Nebius when present	OpenAI model access
OPENAI_MODEL	gpt-4o-mini	OpenAI synthesis model
OPENAI_ROUTER_MODEL	gpt-4o-mini	OpenAI router/research/analysis
OPENAI_BASE_URL	Empty (official endpoint)	Optional OpenAI-compatible base URL
NEBIUS_API_KEY	Optional alternative model provider	Nebius model access
NEBIUS_MODEL	Qwen/Qwen3-30B-A3B-Instruct-2507	Nebius synthesis model
NEBIUS_ROUTER_MODEL	Qwen/Qwen3-30B-A3B-Instruct-2507	Nebius router/research/analysis
NEBIUS_BASE_URL	https://api.studio.nebius.ai/v1/	OpenAI-compatible Nebius endpoint
SCOUT_ROUTER_MODEL	Unset	Provider-independent router override
TAVILY_API_KEY	Required for live mode	Web research
LANGSMITH_API_KEY	Optional	Nested tracing
LANGSMITH_PROJECT	scout-ai	Trace project routing
LANGSMITH_TRACING	true when LangSmith key exists	Tracing enablement

15.3 User-Selectable Parameters

Control	Values	Effect
Depth	quick | comprehensive | exhaustive	Prompt/tag metadata; future policy may vary search and synthesis budgets
Focus	general | finance | news | academic	Brief and trace context; web tool natively supports general/news/finance
Final model	Configured OpenAI/Nebius selections	Writer synthesis model
Routing model	Fast provider-specific model	Orchestrator, Research, and Analysis model
Simulation	off by default; explicit toggle	Unverified workflow preview with no live provider requirement

16. Non-Functional Requirements

Category	Target Requirement	Current Baseline / Gap
Availability	99.5% monthly for production service	Local single process; no HA
Durability	No terminal task loss on deploy/restart	In-flight work fails on restart; no checkpoint
Performance	P50 < 180s; P95 < 360s comprehensive	Provider-dependent; telemetry available
Streaming	Terminal UI state <= 2s after backend transition	SSE works while connected; disconnect reconciliation gap
Scalability	Independent API/workers; queue-based horizontal scaling	BackgroundTasks in web process
Security	Auth, RBAC, encrypted secrets, CSRF controls, audit log	No auth; .env key updates; local-use assumption
Privacy	No secrets or sensitive queries in client logs/traces without policy	Keys not returned; query/trace retention requires governance
Accessibility	WCAG 2.1 AA for primary workflow	High contrast improvements; formal audit pending
Auditability	Every decision traceable to evidence IDs and trace run	Artifact IDs and optional LangSmith supported
Testability	Unit, contract, integration, browser, and evaluation gates	Python unit tests exist; browser/eval suite should expand
Compatibility	Current Chrome/Edge/Firefox; responsive desktop/mobile	CDN-based modern browser UI
Maintainability	Version-pinned dependencies and documented config	Python versions identified; frontend CDN versions partly unpinned

17. Use Cases

Use Case	Example Query / Actor	Expected Outputs
Consumer product recommendation	Buyer: 'Find the top three current cars for reliability, fuel economy, resale value, and TCO.'	Current-market evidence ledger; candidate comparison; recommendation; freshness caveats; cited final report
Enterprise vendor strategy	Executive: 'Compare Microsoft, Google, and Amazon enterprise generative-AI strategies.'	Competitive evidence, SWOT/horizons, strategic radar, decision actions
Technology architecture selection	CTO: 'Compare DeepAgents, LangGraph, CrewAI, and AutoGen for production.'	Feature/operating trade-offs, risks, fit-by-use-case, recommendation
Emerging technology timeline	Investor/analyst: 'Assess solid-state battery commercialization from 2026-2030.'	Dated milestones, company evidence, uncertainty ranges, horizon matrix
Supply-chain resilience	Operations leader: 'Evaluate 2nm GAA yields, packaging capacity, and geopolitical exposure.'	Evidence matrix, bottlenecks, scenarios, mitigations
Market intelligence	Product manager: 'Research current competitors, pricing, and customer signals.'	Structured source dossier, comparison, gaps, launch implications
Policy/standards landscape	Compliance analyst: 'Summarize current requirements and implications.'	Primary-source emphasis, effective dates, uncertainty, and verification notes
Workflow preview	Operator without live keys enables Simulation	Clearly labeled research/analysis/report plan with zero verified claims
Post-run audit	Reviewer opens history and artifact tabs	Exact brief, evidence IDs/URLs, analysis lineage, report, trace metadata

17.1 Use-Case Acceptance Pattern

41. The final report states the exact original question and directly answers it.
42. Recommendations link to evidence records and disclose missing or stale data.
43. Consumer-product answers explicitly distinguish current availability from historical model-year evidence.
44. Unsupported metrics, SWOT items, radar values, or confidence values remain absent rather than populated with placeholders.
45. Failures terminate visibly and name the responsible agent when known.

18. Phased Release Plan

Phase	Scope	Exit Criteria
Phase 0 - Current baseline	Local FastAPI application; multi-agent artifacts; event streaming; retries; artifact metrics; history/export; LangSmith	Existing regression suite passes; documented local operation
Phase 1 - Reliability	SSE reconnect/final fetch; task cancellation; graceful deploy drain; durable checkpoint; resumable task; per-stage deadlines	No stale progress after disconnect; restart test resumes or cleanly terminates every task
Phase 2 - Evidence quality	Primary-source policy; date/availability validator; duplicate-source controls; full-text extraction; source snapshots; calibrated eval datasets	Citation-support and freshness thresholds pass benchmark suite
Phase 3 - Production platform	Database, queue workers, auth/RBAC, encrypted secret store, tenant isolation, rate limits, retention policy	Security review and load/SLO tests pass
Phase 4 - Enterprise optimization	Model routing experiments, caching, provider failover, cost budgets, admin/eval dashboards, API versioning	Cost/latency targets met without groundedness regression

19. Risks & Mitigations

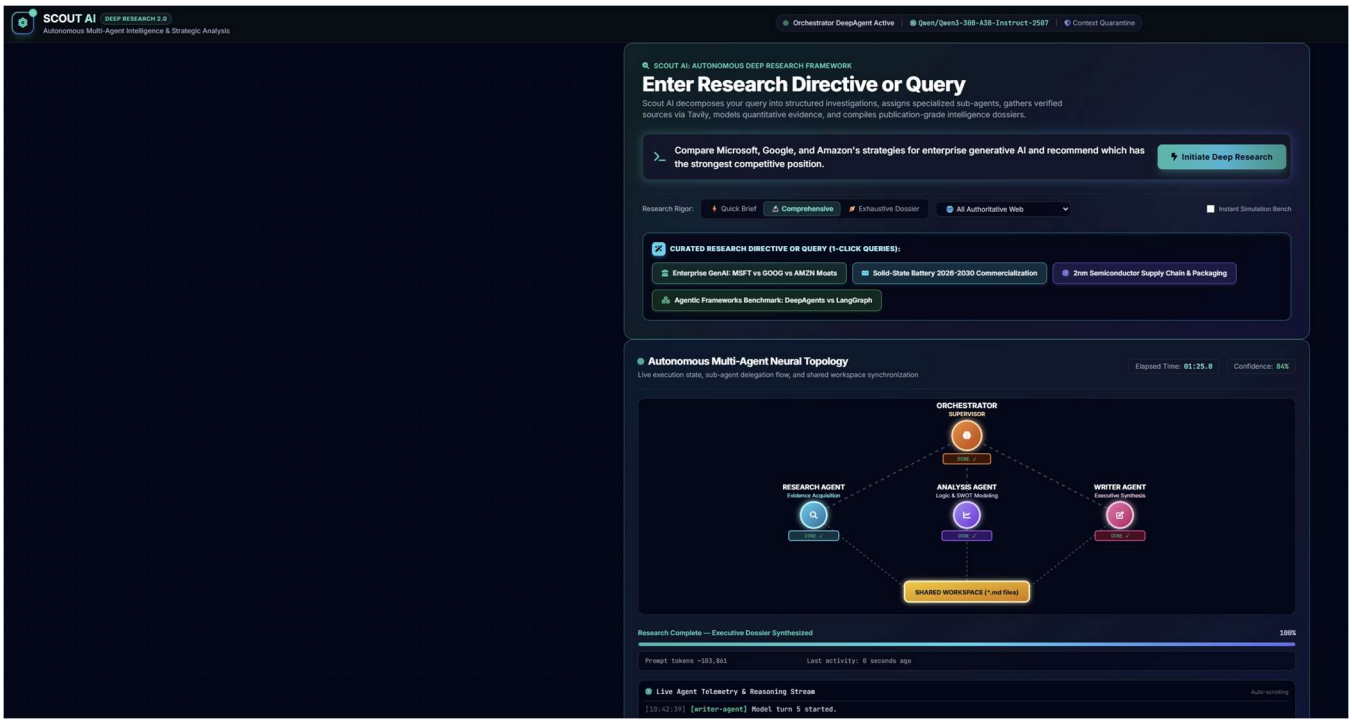
Risk	Impact	Mitigation / Requirement
Application restart during live research	Task appears stalled or fails; work lost	Durable checkpointer and worker queue; graceful drain; UI terminal-state refetch
Provider latency after large search batches	60s inactivity retries, repeated cost, no artifact	Earlier synthesis/write, smaller batches, streaming providers, stage-specific timeout, cache search results across retries
Repeated retries duplicate searches	Token/cost growth and rate-limit pressure	Persist evidence batch and resume synthesis; idempotency keys; retry only failed step
SSE disconnect leaves stale progress	User sees 35% after backend failure	Reconnect with exponential backoff; GET current task on error; render terminal state
JSON persistence under multiple workers	Lost updates or corruption	Transactional database with optimistic locking
Search recency mistaken for product currency	Old model-year recommendation	Validate official availability, on-sale status, publication date, and replacement generation
Snippet-only evidence	Claim misses caveat or methodology	Fetch primary page content; archive excerpt and retrieval timestamp
Prompt/model changes cause loop behavior	Iteration or recursion failures	LangSmith regression evaluations and versioned prompts/models

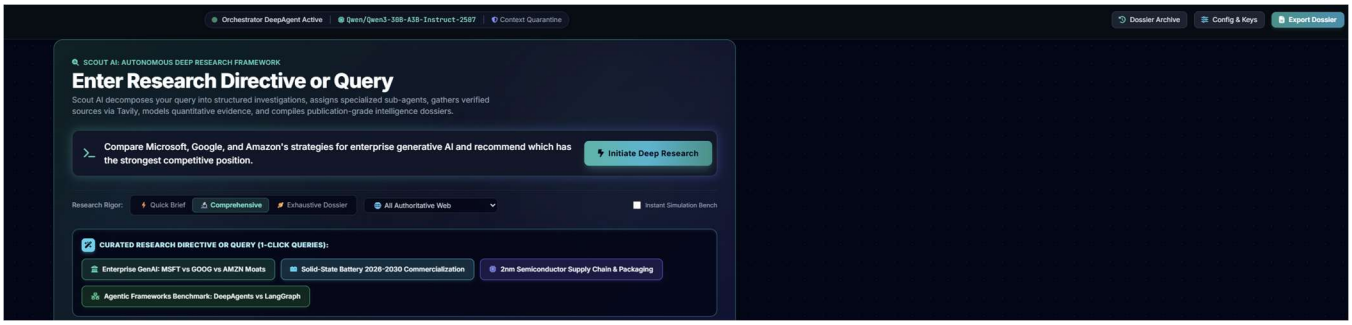
Risk	Impact	Mitigation / Requirement
Settings endpoint stores secrets without auth	Credential compromise	Authentication, secret vault, CSRF protection, no plaintext .env writes in production
Unpinned CDN dependencies	Supply-chain or visual regressions	Pin versions, integrity hashes, or bundle reviewed frontend assets
Confidence score overinterpreted	False certainty	Label as artifact-quality score; expose components and gaps; calibrate against evaluations
No cancellation or queue status	Resource waste and confusing UX	Add queued/running/cancelling/cancelled states and cancellation propagation

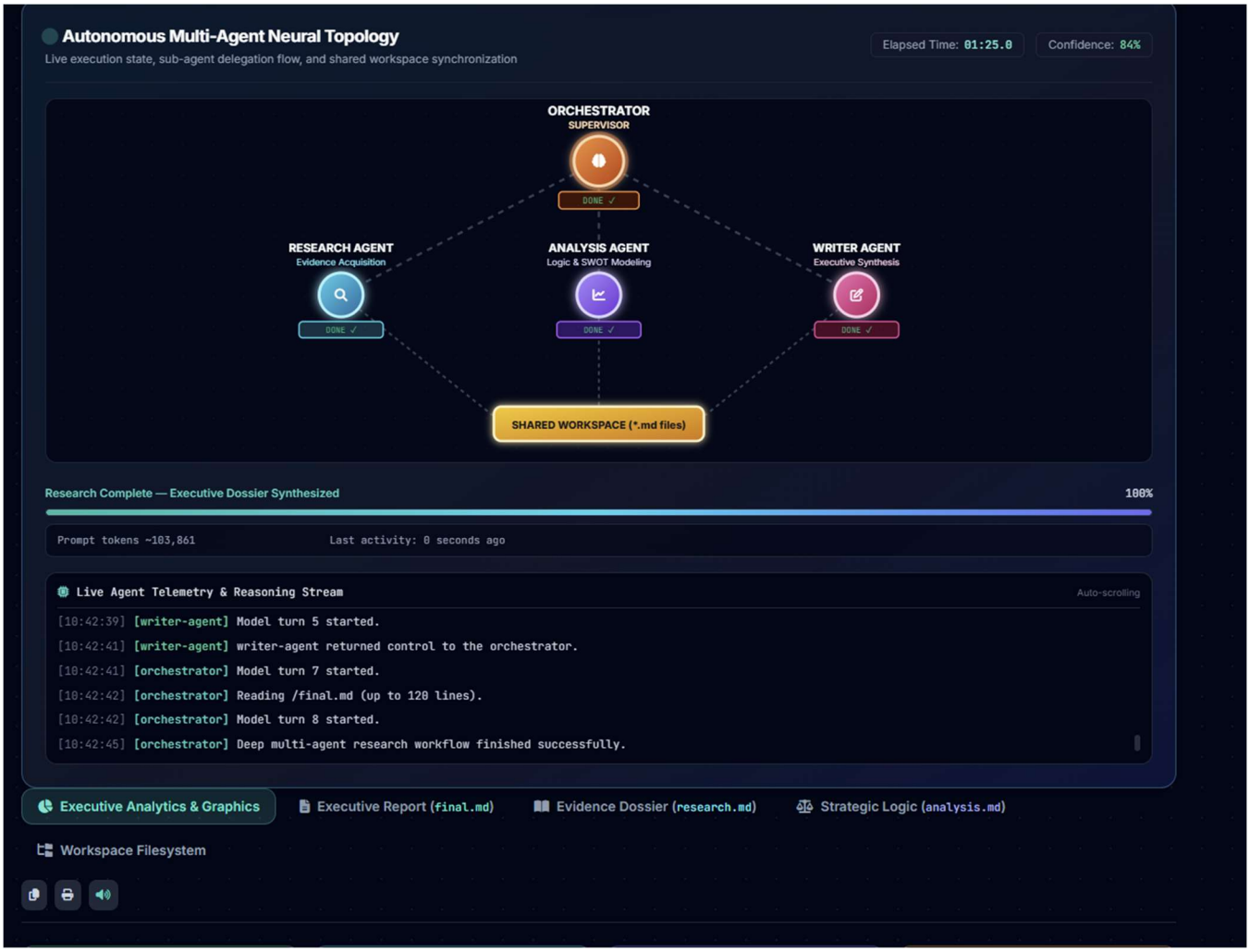
20. Open Questions

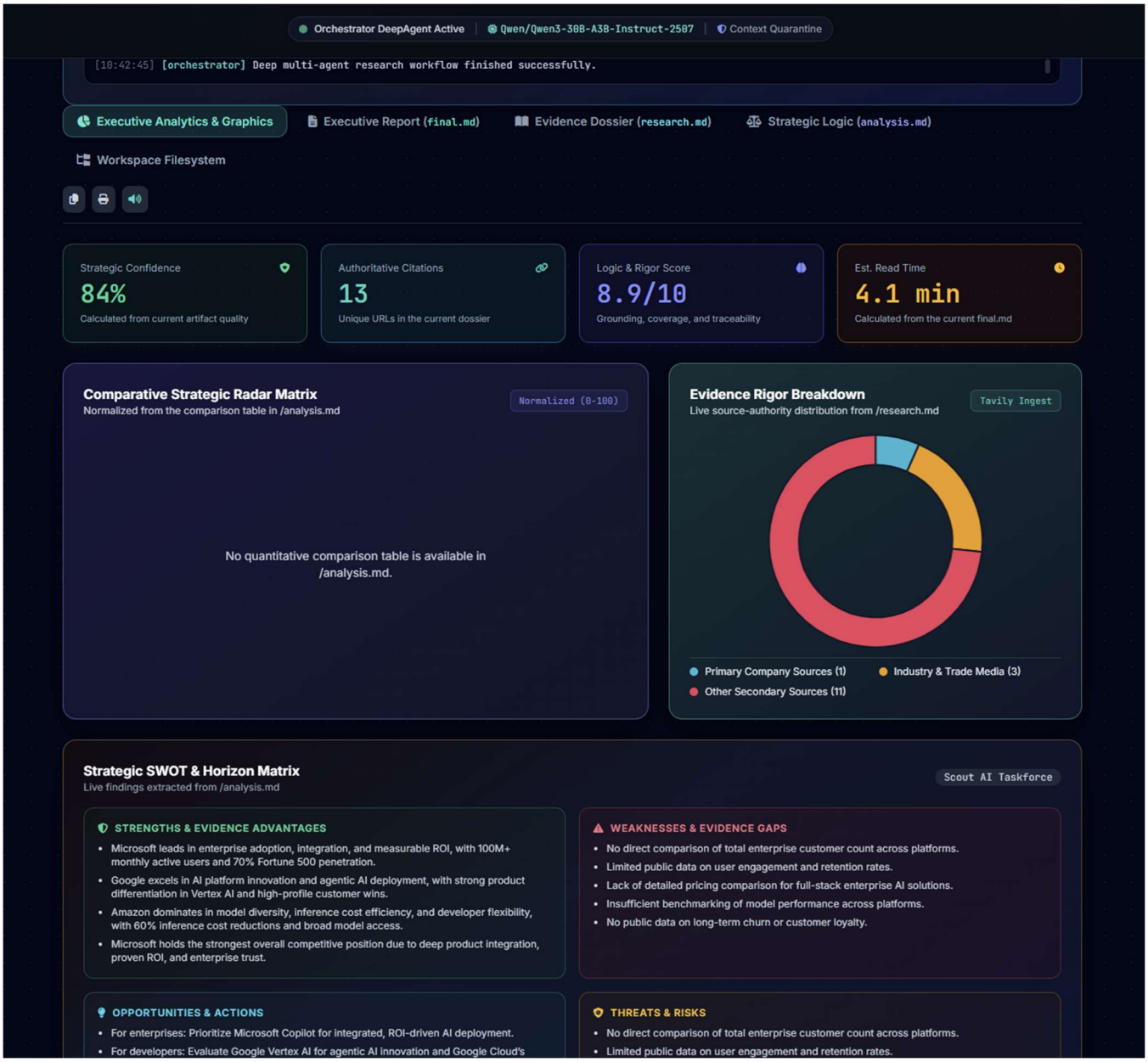
46. Which user segments and research domains define the launch benchmark dataset?
47. Should research depth change agent/search/token budgets, or remain descriptive metadata?
48. Which source domains and source types qualify as authoritative for each focus category?
49. What evidence freshness rule applies when no source exists inside the one-year window?
50. Should the latest-purchasable-product constraint vary by market instead of defaulting to the United States?
51. Which durable execution architecture is preferred: LangGraph checkpointer in the API process, a separate worker queue, or both?
52. What maximum total task cost, token budget, and wall-clock deadline should be enforced?
53. Which models/providers are approved for production, and what failover order applies?
54. What data retention and PII-redaction rules apply to queries, artifacts, JSON/database records, and LangSmith traces?
55. Should users edit and resume intermediate artifacts, or should the pipeline remain fully autonomous?
56. How should the strategic confidence score be calibrated and presented to avoid probability-like interpretation?
57. What authentication, tenant isolation, and audit requirements apply before remote deployment?

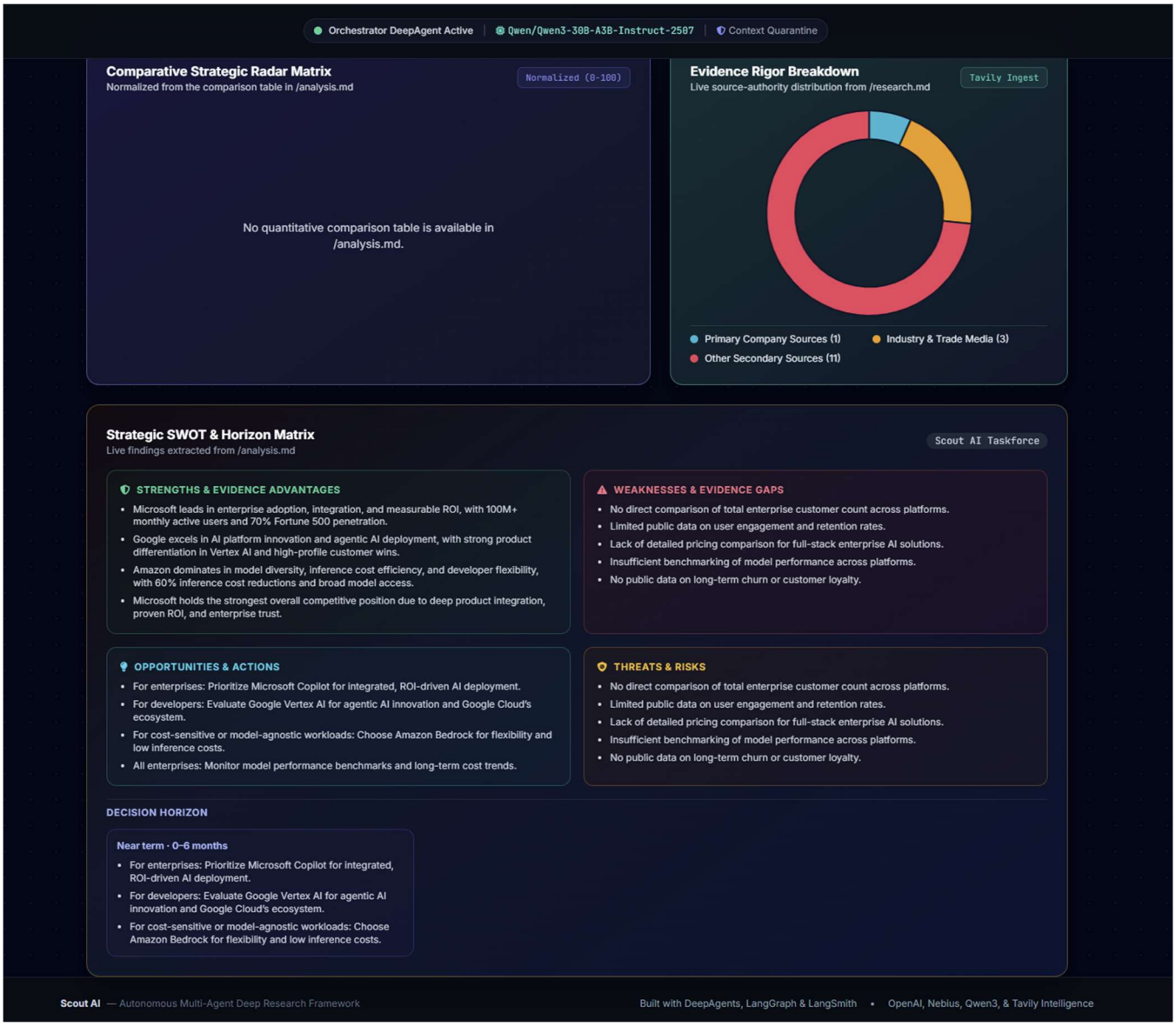
21. User Interface




Orchestrator DeepAgent Active
Qwen/Qwen3-30B-A3B-Instruct-2507
Context Quarantine
Dossier Archive
Config & Keys
Export Dossier
SCOUT AI: AUTONOMOUS DEEP RESEARCH FRAMEWORK
Enter Research Directive or Query
Scout AI decomposes your query into structured investigations, assigns specialized sub-agents, gathers verified sources via Tavily, models quantitative evidence, and compiles publication-grade intelligence dossiers.
Compare Microsoft, Google, and Amazon's strategies for enterprise generative AI and recommend which has the strongest competitive position.
Initiate Deep Research
Research Rigor:
Quick Brief
Comprehensive
Exhaustive Dossier
All Authoritative Web
Instant Simulation Bench
CURATED RESEARCH DIRECTIVE OR QUERY (1-CLICK QUERIES):
Enterprise GenAI: MSFT vs GOOG vs AMZN Moats
Solid-State Battery 2026-2030 Commercialization
2nm Semiconductor Supply Chain & Packaging
Agentic Frameworks Benchmark: DeepAgents vs LangGraph


Autonomous Multi-Agent Neural Topology
Live execution state, sub-agent delegation flow, and shared workspace synchronization
Elapsed Time: 01:25.0
Confidence: 84%
ORCHESTRATOR
SUPERVISOR
DONE ✓
RESEARCH AGENT
Evidence Acquisition
DONE ✓
ANALYSIS AGENT
Logic & SWOT Modeling
DONE ✓
WRITER AGENT
Executive Synthesis
DONE ✓
SHARED WORKSPACE (*.md files)
Research Complete — Executive Dossier Synthesized
100%
Prompt tokens ~103,861
Last activity: 0 seconds ago
Live Agent Telemetry & Reasoning Stream
Auto-scrolling
[10:42:39] [writer-agent] Model turn 5 started.
[10:42:41] [writer-agent] writer-agent returned control to the orchestrator.
[10:42:41] [orchestrator] Model turn 7 started.
[10:42:42] [orchestrator] Reading /final.md (up to 120 lines).
[10:42:42] [orchestrator] Model turn 8 started.
[10:42:45] [orchestrator] Deep multi-agent research workflow finished successfully.
Executive Analytics & Graphics
Executive Report (final.md)
Evidence Dossier (research.md)
Strategic Logic (analysis.md)
Workspace Filesystem


Orchestrator DeepAgent Active
Qwen/Qwen3-30B-A3B-Instruct-2507
Context Quarantine
[10:42:45] [orchestrator] Deep multi-agent research workflow finished successfully.
Executive Analytics & Graphics
Executive Report (final.md)
Evidence Dossier (research.md)
Strategic Logic (analysis.md)
Workspace Filesystem
Strategic Confidence
84%
Calculated from current artifact quality
Authoritative Citations
13
Unique URLs in the current dossier
Logic & Rigor Score
8.9/10
Grounding, coverage, and traceability
Est. Read Time
4.1 min
Calculated from the current final.md
Comparative Strategic Radar Matrix
Normalized from the comparison table in /analysis.md
Normalized (0-100)
No quantitative comparison table is available in /analysis.md.
Evidence Rigor Breakdown
Live source-authority distribution from /research.md
Tavily Ingest
Primary Company Sources (1)
Industry & Trade Media (3)
Other Secondary Sources (11)
Strategic SWOT & Horizon Matrix
Live findings extracted from /analysis.md
Scout AI Taskforce
STRENGTHS & EVIDENCE ADVANTAGES
Microsoft leads in enterprise adoption, integration, and measurable ROI, with 100M+ monthly active users and 70% Fortune 500 penetration.
Google excels in AI platform innovation and agentic AI deployment, with strong product differentiation in Vertex AI and high-profile customer wins.
Amazon dominates in model diversity, inference cost efficiency, and developer flexibility, with 60% inference cost reductions and broad model access.
Microsoft holds the strongest overall competitive position due to deep product integration, proven ROI, and enterprise trust.
WEAKNESSES & EVIDENCE GAPS
No direct comparison of total enterprise customer count across platforms.
Limited public data on user engagement and retention rates.
Lack of detailed pricing comparison for full-stack enterprise AI solutions.
Insufficient benchmarking of model performance across platforms.
No public data on long-term churn or customer loyalty.
OPPORTUNITIES & ACTIONS
For enterprises: Prioritize Microsoft Copilot for integrated, ROI-driven AI deployment.
For developers: Evaluate Google Vertex AI for agentic AI innovation and Google Cloud's
THREATS & RISKS
No direct comparison of total enterprise customer count across platforms.
Limited public data on user engagement and retention rates.


Orchestrator DeepAgent Active
Qwen/Qwen3-30B-A3B-Instruct-2507
Context Quarantine
Comparative Strategic Radar Matrix
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Limited public data on user engagement and retention rates.
Lack of detailed pricing comparison for full-stack enterprise AI solutions.
Insufficient benchmarking of model performance across platforms.
No public data on long-term churn or customer loyalty.
OPPORTUNITIES & ACTIONS
For enterprises: Prioritize Microsoft Copilot for integrated, ROI-driven AI deployment.
For developers: Evaluate Google Vertex AI for agentic AI innovation and Google Cloud's ecosystem.
For cost-sensitive or model-agnostic workloads: Choose Amazon Bedrock for flexibility and low inference costs.
All enterprises: Monitor model performance benchmarks and long-term cost trends.
THREATS & RISKS
No direct comparison of total enterprise customer count across platforms.
Limited public data on user engagement and retention rates.
Lack of detailed pricing comparison for full-stack enterprise AI solutions.
Insufficient benchmarking of model performance across platforms.
No public data on long-term churn or customer loyalty.
DECISION HORIZON
Near term · 0–6 months
For enterprises: Prioritize Microsoft Copilot for integrated, ROI-driven AI deployment.
For developers: Evaluate Google Vertex AI for agentic AI innovation and Google Cloud's ecosystem.
For cost-sensitive or model-agnostic workloads: Choose Amazon Bedrock for flexibility and low inference costs.
Scout AI — Autonomous Multi-Agent Deep Research Framework
Built with DeepAgents, LangGraph & LangSmith • OpenAI, Nebius, Qwen3, & Tavily Intelligence

Appendix A - API & Event Contracts

A.1 HTTP Endpoints

Method	Path	Purpose
GET	/	Serve the Scout AI web interface
GET	/api/config	Return provider readiness, model roles, LangSmith status, and model catalog
POST	/api/config	Persist supplied provider/model/LangSmith configuration
POST	/api/research/start	Create a live or simulation task and return task_id
GET	/api/research/stream/{task_id}	Stream current task telemetry and new log entries via SSE
GET	/api/research/history	List dossier summaries in reverse chronological order
GET	/api/research/history/{task_id}	Load a full dossier; recompute live artifact metrics when complete
DELETE	/api/research/history/{task_id}	Delete a stored dossier
POST	/api/export/docx	Generate a Word report with optional research/analysis appendices

A.2 Start Request

```
{
  "query": "string (required, non-empty)",
  "depth": "comprehensive",
  "focus": "general",
  "model": "optional model ID",
  "force_simulation": false
}
```

A.3 SSE Payload Fields

Group	Fields
Identity/status	id, status, progress, mode, error
Activity	active_agent, current_tool, last_activity_at, last_activity_epoch, activity_state
Limits	model_turns, model_turn_limits, graph_recursion_limit, max_attempts, timeout_seconds
Retries	attempts, attempt_started_epochs, backoff_seconds, delegation_counts
Usage	token_usage totals and per-agent breakdown
Quality	sources_count, confidence_score, duration_seconds, router_model, synthesis_model
Incremental output	logs since prior event; current research/analysis/final lengths
Terminal-only	full_files and metrics

Appendix B - Shared Workspace & Prompt Contracts

B.1 Research Ledger

```
Original research question: <exact question>
Scope/sub-questions: 3-6 bullets

| ID | Claim or metric | Date/scope | Source URL | Source type | Confidence |
| E1 | ... | ... | https://... | Primary/secondary | High/medium/low |

Evidence gaps: <= 5 bullets
```

B.2 Analysis Artifact

```
Original research question: <exact question>
Decision summary: <= 5 bullets
Finding matrix: Finding | Evidence IDs | Fact/Inference/Assumption | Decision implication
Compact comparison table
Risks and gaps: <= 5 bullets
Recommended actions: <= 5 bullets
Strategic SWOT & Horizon Matrix with Evidence IDs
```

B.3 Final Report

58. Executive Summary
59. Key Empirical Findings & Comparative Breakdown
60. Quantitative Metrics & Decision Matrix
61. Strategic Recommendations & Action Items
62. Verified Sources & Citations

Appendix C - Glossary

Term	Definition
DeepAgents	LangChain-ecosystem framework used to create the supervisor and planning-capable sub-agents.
LangGraph	Graph execution runtime underlying the agent workflow and version-2 event stream.
LangSmith	Tracing and evaluation service for model, tool, chain, and sub-agent runs.
Tavily	Web-search provider used to retrieve current titles, URLs, snippets, dates, and relevance scores.
Router model	Model used by the Orchestrator, Research Agent, and Analysis Agent for planning/tool selection/reasoning.
Synthesis model	Model reserved for Writer Agent final-report generation.
Virtual workspace	DeepAgents state-backed set of exact shared paths used for inter-stage handoff.
Evidence Ledger	Compact research.md table linking each claim/metric to date/scope, source URL, type, and confidence.
Context quarantine	Isolation in which a sub-agent does not see the original conversation and must read /brief.md.
Artifact budget	Maximum line and character size enforced for an inter-stage Markdown file.
Model-call limit	Per-agent ceiling enforced by ModelCallLimitMiddleware.
Graph recursion limit	Maximum LangGraph node transitions allowed for a run; distinct from model calls.

Term	Definition
Inactivity timeout	Elapsed silence since the most recent model/tool/token/chain event, not total stage duration.
SSE	Server-Sent Events channel used to deliver live task state from FastAPI to the browser.
Strategic confidence	Deterministic score reflecting artifact quality and traceability; not a probability of truth.
Simulation mode	Explicit unverified workflow preview that performs no live evidence research.